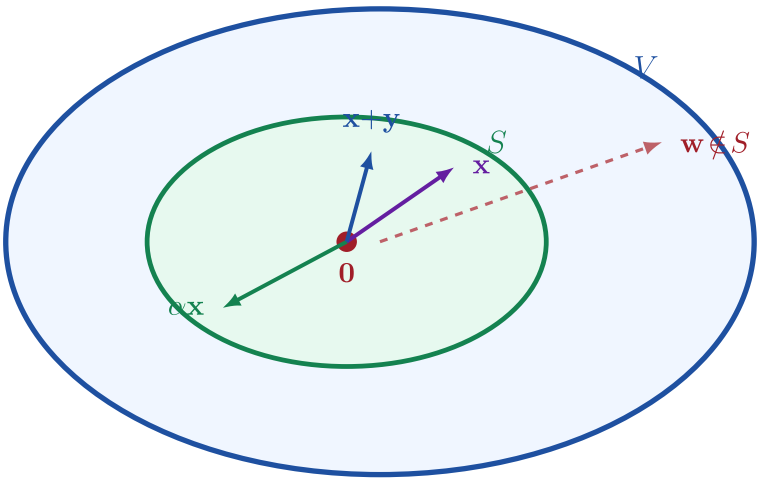
Linear Algebra
Subspaces, Span & Linear Systems

A Comprehensive Mathematical Treatment

Covering subspace theory, null spaces of matrices,
linear combinations, spanning sets, and the
general solution structure of linear systems.



May 21, 2026

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1 Introduction

Having established the abstract notion of a vector space in the previous chapter, we now study the *internal structure* of vector spaces. The central objects are **subspaces** — subsets that are themselves vector spaces under the inherited operations — together with the related notions of **span**, **linear combinations**, and **spanning sets**.

The chapter concludes by applying these ideas to the study of linear systems $A\mathbf{x} = \mathbf{b}$, revealing that the complete solution set has a clean geometric structure built from a particular solution and the null space of A.

2 Subspaces

2.1 Definition and Closure Properties

Definition 2.1: Subspace

Let V be a vector space and let S be a *nonempty* subset of V. S is called a **subspace** of V if it satisfies both of the following *closure conditions*:

(i) **Closed under scalar multiplication:** $\alpha\mathbf{x} \in S$ whenever $\mathbf{x} \in S$, for every scalar $\alpha \in \mathbb{R}$.

(ii) **Closed under addition:** $\mathbf{x} + \mathbf{y} \in S$ whenever $\mathbf{x}, \mathbf{y} \in S$.

Condition (i) ensures that stretching or reflecting any vector in S keeps the result inside S. Condition (ii) ensures that forming the sum of any two vectors in S never "escapes" S. Together, they say that arithmetic with elements of S — using the operations of V — always produces elements of S.

Remark 2.1: Every subspace is itself a vector space

Let S be a subspace of V. Using the operations of V restricted to S, the system $(S, +, \cdot)$ satisfies all eight vector space axioms:

- **A1, A2, A5–A8** hold for any elements of V, so in particular for elements of S.
- **A3** (existence of $\mathbf{0}$): By condition (i), $0 \cdot \mathbf{x} = \mathbf{0} \in S$ for any $\mathbf{x} \in S$ (Theorem 3.1.1(i)).
- **A4** (additive inverse): By condition (i), $(-1)\mathbf{x} = -\mathbf{x} \in S$ for any $\mathbf{x} \in S$ (Theorem 3.1.1(iii)).

Hence S is a vector space in its own right.

2.2 Trivial and Proper Subspaces

Remark 2.2: Trivial subspaces

For any vector space V:

- The **zero subspace** $\{\mathbf{0}\}$ is a subspace of V. It is the smallest possible subspace.
- V itself is a subspace of V. It is the largest possible subspace.

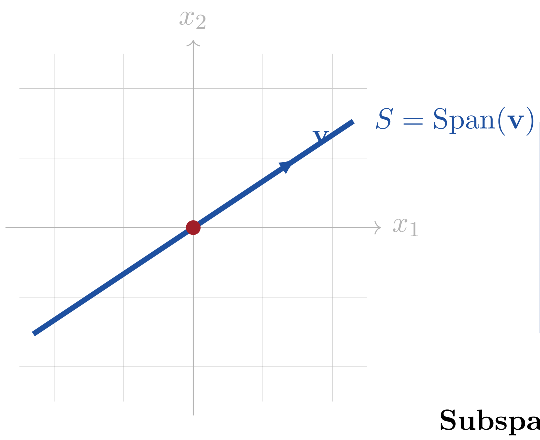
All other subspaces are called **proper subspaces** of V.

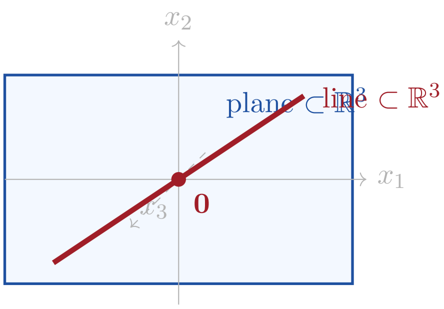
Remark 2.3: Checking nonemptiness via 0

Every subspace must contain the zero vector, since condition (i) with $\alpha = 0$ gives $\mathbf{0} = 0 \cdot \mathbf{x} \in S$. Therefore, the most efficient way to verify that a candidate subset S is nonempty is to check that $\mathbf{0} \in S$.

2.3 Geometric Interpretation in $\mathbb{R}^2$ and $\mathbb{R}^3$

In $\mathbb{R}^2$, the subspaces are exactly: $\{\mathbf{0}\}$, every line through the origin, and $\mathbb{R}^2$ itself. In $\mathbb{R}^3$, the proper subspaces are every line through the origin and every plane through the origin.





Subspaces of $\mathbb{R}^3$: $\{\mathbf{0}\}$, lines, planes, $\mathbb{R}^3$

Subspaces of $\mathbb{R}^2$: $\{\mathbf{0}\}$, lines, $\mathbb{R}^2$

2.4 How to Verify a Subspace

To show that a nonempty subset $S \subseteq V$ is a subspace, we verify exactly three things in order:

for all $\mathbf{x} \in S$, $\alpha \in \mathbb{R}$; [step, right=of s1] (s2) **Step 3**
Show $\mathbf{x} + \mathbf{y} \in S$

for all $\mathbf{x}, \mathbf{y} \in S$; [draw=defgreen, fill=shadegreen, rounded corners=5pt, minimum width=2.5cm, minimum height=1.1cm, align=center, font=**, right=of s2] (s3) S is a subspace; [arr] (s0) – (s1); [arr] (s1) – (s2); [arr] (s2) – (s3);**

3 The Null Space of a Matrix

3.1 Definition

Definition 3.1: Null Space

Let A be an $m \times n$ matrix. The **null space** of A, denoted $N(A)$, is the set of all solutions to the homogeneous linear system $A\mathbf{x} = \mathbf{0}$:

$$N(A) = \{ \mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0} \}.$$

3.2 Proof That $N(A)$ Is a Subspace of $\mathbb{R}^n$

Proposition 3.1: $N(A)$ is a subspace of $\mathbb{R}^n$

For any $m \times n$ matrix A, the null space $N(A)$ is a subspace of $\mathbb{R}^n$.

Proof. We verify the three conditions.

Nonemptiness. $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in N(A)$.

Closed under scalar multiplication. Let $\mathbf{x} \in N(A)$ and $\alpha \in \mathbb{R}$. Then:

$$A(\alpha\mathbf{x}) = \alpha A\mathbf{x} = \alpha\mathbf{0} = \mathbf{0},$$

so $\alpha\mathbf{x} \in N(A)$.

Closed under addition. Let $\mathbf{x}, \mathbf{y} \in N(A)$. Then:

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0},$$

so $\mathbf{x} + \mathbf{y} \in N(A)$.

All three conditions hold; therefore $N(A)$ is a subspace of $\mathbb{R}^n$. ■

3.3 Worked Example

Example 3.1: Computing $N(A)$ for a 2×4 matrix

Let

$$A = \begin{pmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -1 & 3 \end{pmatrix}.$$

Row-reducing the augmented matrix $[A \mid \mathbf{0}]$:

$$\begin{pmatrix} 1 & 2 & -1 & 1 & 0 \\ 2 & 4 & -1 & 3 & 0 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix} \xrightarrow{R_1 + R_2} \begin{pmatrix} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

Setting free variables $x_2 = s,\ x_4 = t$ gives:

$$x_1 = -2s - 2t, \quad x_3 = -t.$$

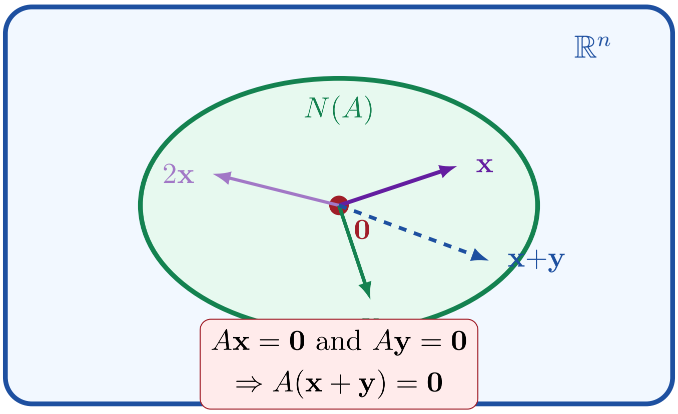
Therefore the general solution is:

$$\mathbf{x} = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \quad s, t \in \mathbb{R},$$

and the null space is

$$N(A) = \operatorname{Span}\left(\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix} \right).$$

3.4 Geometric Picture of the Null Space



4 Linear Combinations and the Span

4.1 Linear Combinations

Definition 4.1: Linear Combination

Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ be vectors in a vector space V. An expression of the form

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n, \quad \alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{R},$$

is called a **linear combination** of $\mathbf{v}_1, \ldots, \mathbf{v}_n$. The scalars $\alpha_1, \ldots, \alpha_n$ are the **coefficients** of the linear combination.

Example 4.1: Linear combination in $\mathbb{R}^3$

Let $\mathbf{v}_1 = (1, 0, 2)^T$, $\mathbf{v}_2 = (0, 1, -1)^T$. Taking $\alpha_1 = 3,\ \alpha_2 = -2$:

$$3\mathbf{v}_1 + (-2)\mathbf{v}_2 = 3\begin{pmatrix}1\\0\\2\end{pmatrix} - 2\begin{pmatrix}0\\1\\-1\end{pmatrix} = \begin{pmatrix}3\\-2\\8\end{pmatrix}.$$

The vector $(3, -2, 8)^T$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

4.2 The Span

Definition 4.2: Span

Let $\mathbf{v}_1, \ldots, \mathbf{v}_n \in V$. The **span** of $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is the set of *all* linear combinations:

$$\mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n) \;=\; \left\{ \alpha_1\mathbf{v}_1 + \cdots + \alpha_n\mathbf{v}_n \;\middle|\; \alpha_1, \ldots, \alpha_n \in \mathbb{R} \right\}.$$

Proposition 4.1: The span is always a subspace

For any vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ in a vector space V, the set $\mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a subspace of V.

Proof. Write $W = \mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$.

Nonemptiness. Taking all $\alpha_i = 0$ gives $\mathbf{0} \in W$.

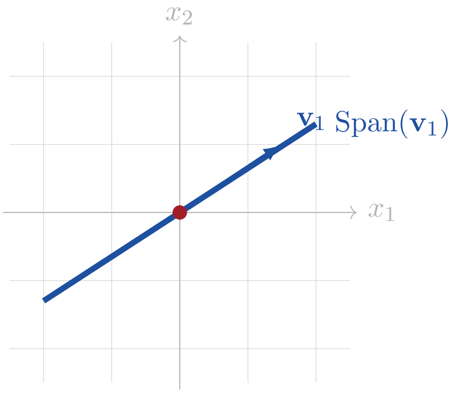
Closed under scalar multiplication. Let $\mathbf{w} = \sum_{i=1}^n \alpha_i\mathbf{v}_i \in W$ and $\beta \in \mathbb{R}$. Then $\beta\mathbf{w} = \sum_{i=1}^n (\beta\alpha_i)\mathbf{v}_i \in W$.

Closed under addition. Let $\mathbf{w} = \sum \alpha_i\mathbf{v}_i$ and $\mathbf{u} = \sum \gamma_i\mathbf{v}_i$ both lie in W. Then $\mathbf{w} + \mathbf{u} = \sum(\alpha_i + \gamma_i)\mathbf{v}_i \in W$.

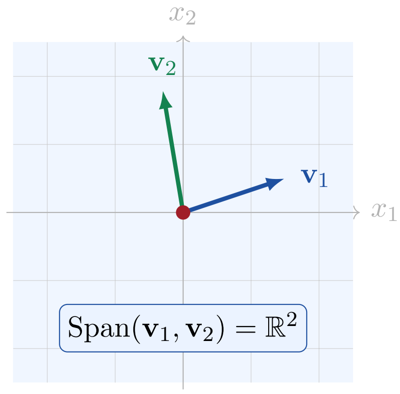
Hence W is a subspace of V. ∎

4.3 Geometric Illustration of Span in $\mathbb{R}^3$

The following diagrams show how the span grows as more vectors are included.



Span$(\mathbf{v}_1)$: **a line**



Span$(\mathbf{v}_1, \mathbf{v}_2)$: **a plane**

5 Spanning Sets for a Vector Space

Definition 5.1: Spanning Set

The set $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a **spanning set for** V if and only if every vector in V can be written as a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$; equivalently,

$$\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_n) = V.$$

Example 5.1: Standard spanning set for $\mathbb{R}^3$

Let

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

For any $(x_1, x_2, x_3)^T \in \mathbb{R}^3$:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3.$$

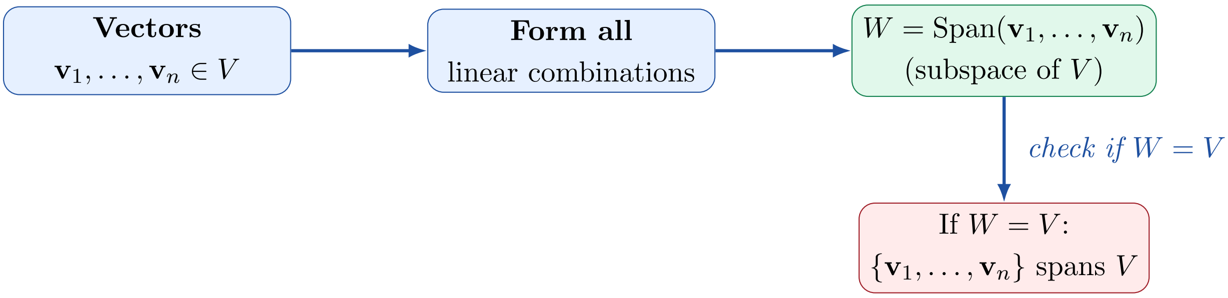
Therefore $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ spans $\mathbb{R}^3$.

Example 5.2: Spanning set for P_3

The set $\{1, x, x^2\}$ spans P_3 because every polynomial of degree less than 3 can be written as:

$$p(x) = a_0 \cdot 1 + a_1 \cdot x + a_2 \cdot x^2, \quad a_0, a_1, a_2 \in \mathbb{R}.$$

The relationship between a vector space V, a spanning set, and the span is illustrated below.



6 Solution Structure of Linear Systems

6.1 The Homogeneous Case: $A\mathbf{x} = \mathbf{0}$

When $\mathbf{b} = \mathbf{0}$, the solution set is $S = N(A)$, which — as proved in Section **??** — is a subspace of $\mathbb{R}^n$. In particular, it contains $\mathbf{0}$, and any linear combination of solutions is again a solution.

6.2 The Nonhomogeneous Case: $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$

When $\mathbf{b} \neq \mathbf{0}$, the solution set is *not* a subspace (it does not contain $\mathbf{0}$ unless $\mathbf{0}$ is a solution, which would require $\mathbf{b} = \mathbf{0}$). Nevertheless, it has a clean structure built from the null space.

Theorem 6.1: General Solution of a Consistent System (Theorem 3.2.2)

Let $A\mathbf{x} = \mathbf{b}$ be a consistent $m \times n$ linear system and let $\mathbf{x}_0$ be a particular solution. Then

$$\boxed{\mathbf{y} \text{ is a solution to } A\mathbf{x} = \mathbf{b} \iff \mathbf{y} = \mathbf{x}_0 + \mathbf{z}, \quad \mathbf{z} \in N(A).}$$

In other words, the complete solution set is

$$S \;=\; \mathbf{x}_0 + N(A) \;=\; \left\{ \mathbf{x}_0 + \mathbf{z} \;\middle|\; \mathbf{z} \in N(A) \right\}.$$

Proof. ($\Leftarrow$) Let $\mathbf{y} = \mathbf{x}_0 + \mathbf{z}$ with $\mathbf{z} \in N(A)$. Then:

$$A\mathbf{y} = A(\mathbf{x}_0 + \mathbf{z}) = A\mathbf{x}_0 + A\mathbf{z} = \mathbf{b} + \mathbf{0} = \mathbf{b}.$$

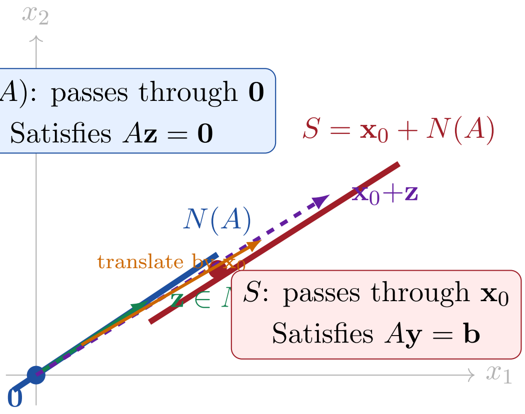
($\Rightarrow$) Let $\mathbf{y}$ be any solution, so $A\mathbf{y} = \mathbf{b}$. Set $\mathbf{z} = \mathbf{y} - \mathbf{x}_0$. Then:

$$A\mathbf{z} = A\mathbf{y} - A\mathbf{x}_0 = \mathbf{b} - \mathbf{b} = \mathbf{0},$$

so $\mathbf{z} \in N(A)$, giving $\mathbf{y} = \mathbf{x}_0 + \mathbf{z}$. ■

6.3 Geometric Interpretation

The solution set $S = \mathbf{x}_0 + N(A)$ is a *translate* (or coset) of the null space. When $N(A)$ is a line through the origin in $\mathbb{R}^n$, the solution set is a parallel line passing through $\mathbf{x}_0$. When $N(A)$ is a plane, S is a parallel plane, and so on.



6.4 Worked Example

Example 6.1: Complete solution of $A\mathbf{x} = \mathbf{b}$

Let

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 8 \end{pmatrix}.$$

Step 1: Find a particular solution. Row-reduce $[A \mid \mathbf{b}]$:

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 8 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_1 - R_2} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}.$$

Setting the free variable $x_2 = 0$: $x_1 = 1$, $x_3 = 2$. A particular solution is $\mathbf{x}_0 = (1, 0, 2)^T$.

Step 2: Find $N(A)$. With $x_2 = s$ free and $\mathbf{b} = \mathbf{0}$: $x_1 = -2s$, $x_3 = 0$.

$$N(A) = \operatorname{Span}\left(\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \right).$$

Step 3: Write the complete solution.

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{z} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \quad s \in \mathbb{R}.$$

7 Summary and Connections

Chapter Summary

- A **subspace** S of a vector space V is a nonempty subset closed under scalar multiplication and addition; it is itself a vector space.
- Every subspace contains $\mathbf{0}$; use this to check nonemptiness.
- The **trivial subspaces** of V are $\{\mathbf{0}\}$ and V itself.
- The **null space** $N(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$ is always a subspace of $\mathbb{R}^n$.
- A **linear combination** of vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ is any sum $\alpha_1\mathbf{v}_1 + \cdots + \alpha_n\mathbf{v}_n$.
- The **span** of a set of vectors is the set of all their linear combinations; it is always a subspace.
- $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a **spanning set** for V when every vector in V is a linear combination of them.
- For a consistent system $A\mathbf{x} = \mathbf{b}$, the general solution is $\mathbf{y} = \mathbf{x}_0 + \mathbf{z}$, where $\mathbf{x}_0$ is any particular solution and $\mathbf{z}$ ranges over $N(A)$.

The following diagram summarises all the key relationships covered in this chapter.